have a favourable limit". In order to formulate the concept of continuity in terms of limit we must focus our attention at a point. Both continuity and limit are primarily concepts defined at a point, but continuity acquires a global character in a pointwise way.

For motivation, consider the following physical situation. A thermometer T measures temperatures along a given hot wire L.

To each point x on the hot wire L is assigned a temperature readings $t(x)$ on thermometer, T. Suppose, to fix ideas, the temperature recordings are observed to be the same, 250°F, as one moves along the wire L until a point x_0 is reached on L.

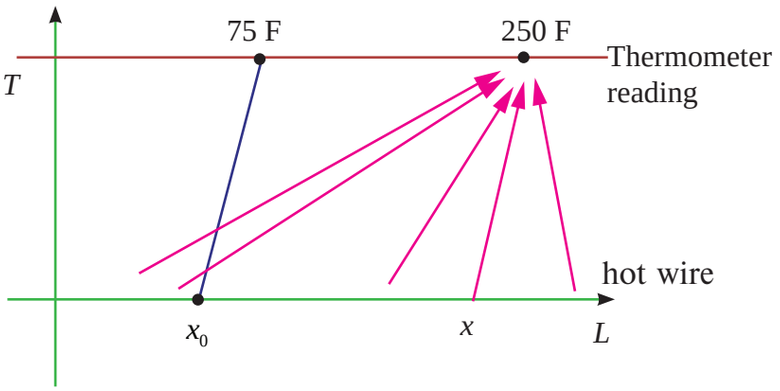


Fig. 9.32

Then suppose that at x_0 the temperature drops suddenly to near room temperature, say 75°F, as if an insulation were at x_0. But, beyond x_0 suppose readings of 250°F are again observed. In function notation we are assuming that

$$t(x) = \begin{cases} 250^\circ F & \text{if } x \neq x_0 \\ 75^\circ F & \text{if } x = x_0 \end{cases}$$

Thus the point x_0 stands out as singular point ("singular" means "special" or "unusual"). Analyzing the range of temperature readings, we should say that the approach of x to x_0 had no bearing on the approach of the corresponding $t(x)$ values to $t(x_0)$. Briefly, a jump occurs at x_0. Thus we would be led to say that the temperature function "lacks continuity at x_0". For, we would have expected $t(x_0)$ to be 250°F since the x values neighbouring x_0 showed $t(x)$ = 250°F. We now abstract the notion of continuity, and demand that images be "close" when pre-images are "close". In our example the points on the hot wire were the pre-images, while the temperature readings there were the corresponding images.

The students should reflect on the intuitive idea of continuity by considering instead the contrasting idea of lack of continuity, or more simply "discontinuity" as manifested in every day experiences of abrupt changes which could be headed "then suddenly!". A few that come readily to mind are listed below along with functions which correspond as mathematical models:

(1) Switching on a light : light intensity as a function of time.

(2) Collision of a vehicle : Velocity as a function of time.

(3) Switching off a radio : Sound intensity as a function of time.

(4) Busting of a balloon: Radius as a function of air input.

(5) Breaking of a string : Tension as a function of length.

(6) Cost of postage : Postage as a function of weight.

(7) Income tax : Tax-rate as a function of taxable income.

(8) Age count in years : Age in whole years as a function of time.

(9) Cost of insurance premium : Premium as a function of age.

Actually, examples (1) – (5) are not quite accurate. For example, light intensity has a transparent but continuous passage from zero luminosity to positive luminosity. Indeed, nature appears to abhor

09_11th-Maths-Vol2-Unit9.indd 119 16-12-2022 11:03:41